

Problem Solving

1. Let $X_1, \dots, X_n$ denote a random sample from $N(\mu, \theta)$, $0 < \theta < \infty$, where μ is unknown. Let $Y = n^{-1} \sum_{i=1}^n (X_i - \bar{X})^2$ and let $L(\theta, \delta(y)) = (\theta - \delta(y))^2$. If we consider decision functions of the form $\delta(y) = b y$, where b does not depend on y . Show that $b = n/(n+1)$ yields a minimum risk decision function of this form. Note that $nY/(n+1)$ is not an unbiased estimator of θ . With $\delta(y) = ny/(n+1)$, determine $\max_{\theta} R(\theta, \delta)$ if it exists.

2. Let $X_1, \dots, X_n$ be iid $N(0, \theta)$, $0 < \theta < \infty$. What is the sufficient statistic for θ ?

3. Let $X_1, \dots, X_n$ be iid $Unif(-\theta, \theta)$, $0 < \theta < \infty$. What is the sufficient statistic for θ ?

4. Let $X_1, \dots, X_n$ be a random sample from a Poisson distribution with mean θ . Find the conditional expectation $\mathbb{E}(X_1 + 2X_2 + 3X_3 | \sum_{i=1}^n X_i)$.

